Vinny Vinh Quoc Pham

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Education

De Anza Community College — Cupertino, CA

A.S. Computer Science, Transfer Track — *Sep. 2024 – Expected May 2027*

Associate in Arts for Transfer, Business Administration (AA-T) — *Expected Winter 2026*

Projects

CacheProof | GitHub | *Python, FastAPI, MCP, pgvector* — Beta Fund × Snowflake × Evermind, Aug. 2026

- Built an AST-aware semantic cache for the read-only repository tools used by cloud coding agents, hard-scoping every lookup by tenant, repository, commit, permission hash, and tool schema so cached results cannot cross a trust boundary
- Reduced median retrieval from 2,973 to 161 tokens (95% smaller) by serving AST-scoped symbol slices instead of full files, and eliminated wrong-symbol cache hits that a file-level cache produced on a near-duplicate-symbol stress fixture
- Exposed the cache over FastAPI and an MCP stdio server with a fail-closed LLM judge that returned 0 false approvals across 40 adversarial symbol pairs
- Wrote a deterministic replay harness and dashboard that meter retrieval, embedding, judging, and indexing cost per call, plus a test suite covering scope isolation, commit invalidation, and cache-decision logic

Codebase Search Engine | GitHub | *React, TypeScript, Vite, Neon, pgvector, OpenAI API*

- Built a semantic code search tool with one other developer that indexes GitHub repositories and lets developers search their code using natural language instead of exact keywords
- Integrated the OpenAI API to generate embeddings for indexed code, enabling concept-based search that matches how developers actually remember code
- Stored and queried vector embeddings using pgvector on Neon (serverless Postgres) for fast similarity search across repositories

ProcureMate | GitHub | *React, TypeScript, Vite, FastAPI, SQLAlchemy, PostgreSQL*

- Built a full-stack async procurement app using FastAPI, PostgreSQL, and Redis with multi-layered auth (OAuth, sessions, API keys) and configurable rate limiting
- Implemented background job processing with Taskiq and generic CRUD with FastCRUD

Experience

IT Technician Volunteer — May 2020 – Mar. 2025

Lamson Manufacturing

- Repaired and maintained company computers, printers, networking equipment, and office devices
- Provided fast IT support under tight deadlines to keep systems running efficiently

Tea Barista / Team Member — Oct. 2025 – Feb. 2026

Rabbit Rabbit Tea

- Operated POS systems, prepared beverages, and collaborated with team members to maintain efficient workflow

Technical Skills

Languages: C++, Java, JavaScript, TypeScript, Python, HTML/CSS
Frontend: React, Next.js, TanStack Start, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, PostgreSQL
Developer Tools: Git, GitHub, Arduino, Claude Code, Codex, WebStorm, PyCharm, Visual Studio Code, Supabase, Neon